

UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS
FACULTAD DE INGENIERÍA DE SISTEMAS E INFORMÁTICA
DOCTORAL PROGRAM IN DEEP TECH, ARTIFICIAL INTELLIGENCE & EMERGING
TECHNOLOGIES



RESEARCH METHODS AND SCIENTIFIC INTEGRITY IN AI AND ADVANCED
TECHNOLOGIES

Deliverable:

Protocol Outline v0.1

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1. Title

Explainable Artificial Intelligence for Early Prediction of School Interruption Risk among Secondary Education Students in Peru

2. Abstract

The proposed research addresses the early identification of school interruption risk among secondary education students in Peru through the design and rigorous evaluation of an Explainable Artificial Intelligence artifact. Using Design Science Research as the governing approach, the study will examine predictive performance together with calibration, explanation faithfulness and stability, fairness across relevant groups, reproducibility, and safeguards for responsible human use.

3. Introduction and Problem Statement

School interruption is a relevant challenge for the Peruvian educational system because continued participation in secondary education may be affected by interacting academic, socioeconomic, demographic, family, institutional, and territorial factors. Although Peru already has early-warning initiatives such as *Alerta Escuela*, the research problem is not merely whether school interruption can be predicted, but whether an AI-based early-warning artifact can provide predictions that are sufficiently accurate, calibrated, explainable, stable, fair across relevant groups, and suitable for responsible human decision support.

The specific research gap will be confirmed through the systematic literature review in Session 4. In particular, the review will determine whether existing research in the Peruvian or comparable educational context has jointly evaluated predictive performance, explanation faithfulness and stability, fairness, temporal validity, and responsible educational use.

4. Literature Review

The literature review will examine four connected bodies of knowledge: early-warning and school-interruption prediction in education; Machine Learning models for student-risk prediction; Explainable Artificial Intelligence methods used in high-stakes decision-support systems; and research on fairness, stability, calibration, reproducibility, and human oversight in AI.

A systematic search will subsequently be designed using the PRISMA 2020 framework to determine the documented research gap and position the proposed artifact against existing educational early-warning approaches, including the Peruvian context.

5. Research Questions and Objectives

General Research Question

RQ1. How can an Explainable Artificial Intelligence artifact be designed and rigorously evaluated to support the early identification of school interruption risk among secondary education students in Peru?

Specific Research Questions

RQ2. Which predictive modeling approaches provide the most robust performance, calibration, and temporal generalization for identifying students at risk of school interruption?

RQ3. To what extent are the explanations generated by the selected XAI approach faithful, stable, and fair across relevant territorial and demographic groups?

General Objective

To design and rigorously evaluate an Explainable Artificial Intelligence artifact for the early identification of school interruption risk among secondary education students in Peru.

Specific Objectives

O1. Identify and operationalize the educational, demographic, socioeconomic, and territorial variables that can be legitimately used to model school interruption risk.

O2. Develop and compare supervised Machine Learning models using reproducible evaluation procedures, including appropriate baselines, calibration, class-imbalance handling, and temporal validation where data permit.

O3. Integrate and evaluate Explainable AI mechanisms by examining explanation faithfulness, stability, and fairness across relevant groups.

O4. Define a human-in-the-loop early-warning prototype in which predictions and explanations support, rather than replace, professional educational judgment.

6. Methodology

Working methodological outline

Philosophical position: Pragmatism

Primary research approach: Design Science Research

Embedded evaluation strategy: Empirical Machine Learning evaluation

The research will follow a Design Science Research build–evaluate–refine cycle. The artifact will integrate predictive Machine Learning and Explainable AI mechanisms, while empirical evaluation will examine predictive performance, calibration, class imbalance, temporal validity where feasible, reproducibility, explanation faithfulness and stability, and fairness across selected groups.

Data development will depend on obtaining educational data with sufficient student-level granularity and temporal structure. Candidate models may include appropriate interpretable and non-linear baselines, while the final modeling set will be justified from the literature rather than selected solely on expected accuracy.

7. Ethical Considerations

Because the research may involve educational records related to minors, ethics and data governance will be treated as core methodological requirements rather than administrative formalities. Approval from the relevant UNMSM Research Ethics Committee will be obtained before accessing or collecting restricted research data, and the study will incorporate data minimization, de-identification, access control, secure storage, non-reidentification procedures, and appropriate retention/deletion policies.

The protocol will be aligned, where applicable, with the CONCYTEC National Code of Scientific Integrity, Ley N.º 29733 and its current personal-data regulation, and the Peruvian AI regulatory framework. Particular attention will be given to fairness, non-discrimination, transparency, human oversight, and the risk that an AI-generated risk score could stigmatize a student or become a self-fulfilling label.

8. Expected Results

The expected outputs are a rigorously evaluated XAI early-warning artifact; a comparative and reproducible assessment of selected predictive models; an analysis of calibration and temporal generalization; an evaluation of explanation faithfulness and stability; and a fairness assessment across relevant territorial or demographic groups supported by the available data.

The project is also expected to produce reusable design knowledge concerning how predictive performance, explainability, stability, fairness, and human oversight can be jointly considered when developing AI-based educational early-warning systems in heterogeneous contexts such as Peru.

9. Timeline and Budget

10. Preliminary Bibliography

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